In [1]:
```python
import pandas as pd
import matplotlib.pyplot as plt
```

In [3]:
```python
df=pd.read_csv(r"C:\Users\Dell\BostonHousing.csv")
```

In [4]:
```python
df
```

Out[4]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	l
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	
...	...	...	...	...	...	...	...	...	...	...	...	...	
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396.90	

506 rows × 14 columns

In [5]:
```python
df.isnull().sum()
```

Out[5]:
```
crim       0
zn         0
indus      0
chas       0
nox        0
rm         5
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
medv       0
dtype: int64
```

In [6]:
```python
df.shape
```

Out[6]:
```
(506, 14)
```

In [7]:
```python
df['rm']=df['rm'].fillna(df['rm'].mean())
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
crim       0
zn         0
indus      0
chas       0
nox        0
rm         0
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
medv       0
dtype: int64
```

In [10]:
```
x=df.drop('medv',axis=1)
```

In [11]:
```
y=df['medv']
```

In [12]:
```
from sklearn.model_selection import train_test_split
```

In [13]:
```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=0)
```

In [14]:
```
from sklearn.linear_model import LinearRegression
```

In [15]:
```
model=LinearRegression()
```

In [17]:
```
model.fit(x_train,y_train)
```

Out[17]:

▾ LinearRegression

▶ Parameters

In [18]:
```
y_pred=model.predict(x_test)
```

In [19]:
```
print(y_test)
print(y_pred)
```

```
329    22.6
371    50.0
219    23.0
403     8.3
78     21.2
       ...
56     24.7
455    14.1
60     18.7
213    28.1
108    19.8
Name: medv, Length: 102, dtype: float64
[24.87329469 23.72648298 29.37933969 12.14039753 21.4426036  19.29805136
 20.50989946 21.37919491 18.90775641 19.91250289  5.13511986 16.36549318
 17.07820755  5.61485991 39.97664469 32.50308769 22.46740518 36.8539512
 30.84639924 23.15702962 24.77730389 24.68458506 20.57397739 30.34986683
 22.43591012 10.296692   17.65487529 18.29000068 35.48502785 20.94457768
 18.30381077 17.80605213 20.00702001 24.08474181 29.09076024 19.29811977
 11.14558052 24.59253369 17.57876353 15.48429212 26.21883229 20.87030216
 22.31042069 15.62684761 22.99790444 25.18118404 20.14259612 22.91240755
 10.04895192 24.26059172 20.94855315 17.33461914 24.5292998  29.94042972
 13.42890008 21.73824546 20.80915203 15.52227571 14.01162977 22.19280817
 17.78077486 21.58933891 32.89946019 31.10232804 17.74375021 32.75366803
 18.70665918 19.46000209 19.00433221 22.89791727 22.95526653 24.02145361
 30.74669042 28.83499729 25.90273982  5.23616938 36.68379136 23.77290327
 27.25150674 19.29151539 28.63422641 19.16289464 18.9909208  37.78003086
 39.16489139 23.721517   25.36461611 15.88729711 26.10855921 16.68957745
 15.84600328 13.09338048 24.76001983 31.27024775 22.16809052 20.28503266
  0.6038393  25.48075438 15.56795821 17.98050623 25.29830625 22.35187313]
```

In [20]:

```
comparison = pd.DataFrame({
    "Actual Price": y_test.values,
    "Predicted Price": y_pred
})

print(comparison.head())
```

```
   Actual Price  Predicted Price
0          22.6        24.873295
1          50.0        23.726483
2          23.0        29.379340
3           8.3        12.140398
4          21.2        21.442604
```

In [21]:

```
comparison["Difference"] = comparison["Actual Price"] - comparison["Predicted Pr

print(comparison.head())
```

```
   Actual Price  Predicted Price  Difference
0          22.6        24.873295   -2.273295
1          50.0        23.726483   26.273517
2          23.0        29.379340   -6.379340
3           8.3        12.140398   -3.840398
4          21.2        21.442604   -0.242604
```

In [22]:

```
from sklearn.metrics import mean_squared_error
mse=mean_squared_error(y_test,y_pred)
```

In [23]:

```
mse
```

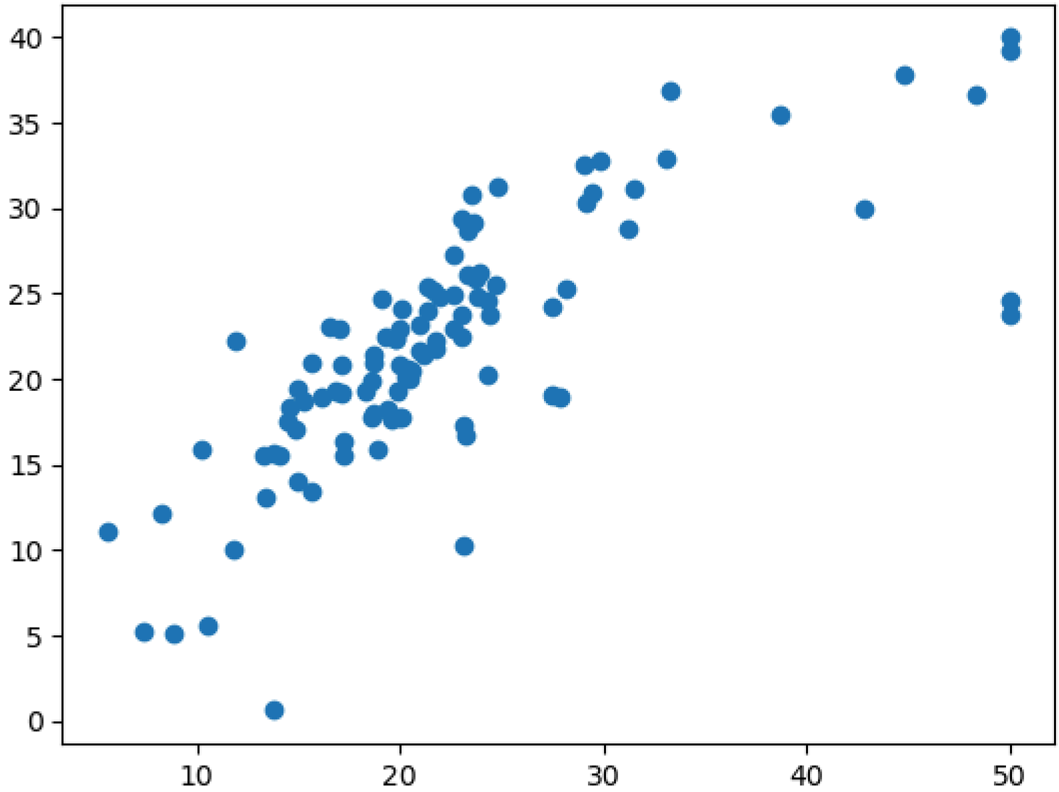
Out[23]: 33.464975988914524

In [25]:
```
from sklearn.metrics import r2_score
r2=r2_score(y_test,y_pred)
r2
```

Out[25]: 0.5890259426012632

In [26]:
```
plt.scatter(y_test,y_pred)
```

Out[26]: <matplotlib.collections.PathCollection at 0x1e9481f3a10>



In [27]:
```
plt.scatter(y_test,y_pred)
plt.xlabel("Actual Prices")
plt.ylabel('Predicted Prices')
plt.title('Actual Prices VS Predicted Prices ')
plt.plot()
plt.show()
```

Actual Prices VS Predicted Prices
Predicted Prices
Actual Prices
0
5
10
15
20
25
30
35
40
10
20
30
40
50

In [ ]: